ORLAITH DOYLE

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PROFILE

Current European Space Agency Intern with degrees in MSc Space Science & Technology and BSc Physics with Astrophysics, along with hands-on space engineering experience across payload design, hardware AIV, CAD and electronics. Led a five-person team through a full balloon-payload mission lifecycle as Project Manager, and currently contributing CAD and technical liaison work on ESA's VPSAT2 satellite project with core focus working on ESA's BIOMASS mission: developing and validating a sea surface salinity retrieval using SAR remote sensing. Broader remote sensing/GIS background spans QGIS, ArcGIS, and project work with SWOT and Sentinel-1/2 data. Open to opportunities in earth observation, space engineering, mission operations, or space science, with availability as early as November 2026.

PROFESSIONAL EXPERIENCE

ESA Remote Sensing Intern, ESTEC Climate Action, Sustainability & Science Dept, EO Missions **05/26 - 11/26**

- Contributed to the Biomass Earth Explorer (EO) mission, supporting scientific and sustainability-related analyses aimed at improving global understanding of the mission secondary objectives including cryospheric and oceanic applications.
- Assisted in analysis and interpretation of EO data to support climate monitoring and carbon accounting initiatives.
- Collaborated within a multi-disciplinary and international team, gaining exposure to satellite missions, environmental modelling, and policy-relevant science including ECSS Standardisations and Review Documents (PDR, CDR, FAR).
- Participated in and supported development of technical reports, presentation, and internal documentation for within ESA.
- Contributed to sustainability and climate action initiatives, aligning scientific outputs with broader environmental objectives and policy frameworks.
- Conducted research on climate science, EO applications, and sustainability practices to inform mission activities.
- Demonstrated ability to translate complex scientific concepts into clear communication for diverse and lay audiences.

Laboratory Technician, DCU School of Physics, Glasnevin, Dublin **08/23 - 04/24**

- Supported optical and physics laboratories through careful experimental setup, data checking, and documentation to ensure reliable and repeatable results.
- Assisted students and staff in understanding experimental methods, instrumentation, and expected outcomes.
- Performed data checks and contributed to the documentation of experimental procedures and results.
- Worked closely with academic staff to maintain clear laboratory instructions and support effective knowledge transfer during practical sessions.

Founder, Private Tutor & Website Developer, Grade+ Online Tutoring & Grinds **02/24 - 10/24**

- Founded and managed a private tutoring service providing one-to-one tuition in Physics and Mathematics at secondary and undergraduate level.
- Structured learning materials and explanations to support different learning styles and levels of technical background.
- Assessed student progress and provided clear, written feedback to support understanding and improvement.
- Developed and maintained a tutoring website to organise resources and streamline communication.

Supervisory Roles; (Retail & Hospitality) Dunnes Stores & Kippure Est, Blessington, Co. Wicklow **05/17 - 01/24**

- Coordinated daily operations, task allocation, and staff scheduling.
- Maintained clear procedures and communication to ensure consistent service and smooth handovers.
- Developed organisational and interpersonal skills in fast-paced environments.

EDUCATION & ACADEMIC EXPERIENCE

MSc Space Science & Technology, University College Dublin, Ireland **09/25 - 09/26**

- **Expected GPA**: First Class Honours (1.1)
- **Key Relevant Modules:** Space Detection Lab, Space Mission Design, Technical Communications, Spacecraft Operations, Satellite Subsystems Lab, Applications of Space Science, Space Sector Professional Skills, Space Environment, SAR & InSAR Remote Sensing, Planetary Geomorphology.
- **Relevant Project: Satellite Subsystems Module; High Altitude Balloon Stratospheric Flight**
  Served as Project Manager for the SQRT (Sub-zero Quantitative Research Team) mission, a five-person team designing and flying a 'TupperSat' payload on a high-altitude stratospheric balloon. The mission investigated the flash evaporation of water under near-space low-pressure and low-temperature conditions, with relevance to cryovolcanic plume behaviour on icy moons. Led project documentation, coordinating across science, structure, software, and AIV (Assembly, Integration & Verification) work packages. Authored the PDR, CDR and FAR documents and managed overall project scheduling, budget tracking, hardware design and team coordination across six structured mission phases from concept through post-flight analysis.

- **Relevant Project: Space Detector Laboratory Module; High-Energy Detectors Characterisation & Calibration**
  Conducted a laboratory-based characterisation of gamma-ray detectors (CdTe, NaI(Tl), and BGO), focusing on calibration, resolution, and efficiency analysis. Processed and analysed spectral data using Python to evaluate detector performance and geometry-dependent effects. Interpreted experimental results and documented performance trade-offs and limitations in structured technical reports, developing experience in translating experimental findings into clear, usable technical knowledge.
- **Relevant Project: Space Sector Professional Skills; ESA-Style Requirements & Trade-Off Analysis Proposal**
  Prepared a structured technical proposal in response to an example ESA-style Invitation to Tender (ITT) for a planetary sample return container recovery mechanism. Analysed mission and system requirements, identified ambiguities affecting verifiability, and proposed clarifications to improve traceability. Conducted high-level trade-off analyses across sensing, autonomy, and capture concepts, and documented design rationale, risks, and validation approaches in a clear, structured technical report aligned with ECSS principles.
- **Relevant Project: Geophysics Elective; Monitoring Volcanic Activity at Long Valley Caldera**
  Analysed real geophysical monitoring datasets from Long Valley Caldera, California, using publicly available USGS data. Examined continuous GPS time series from multiple stations (KRAK, CASA, MINS) spanning approximately 1994 to present to identify ground deformation trends indicative of magma inflation or deflation. Correlated deformation signals with seismic earthquake catalogue data to investigate subsurface volcanic processes. Produced figures, interpreted multi-dataset geophysical signals, and communicated findings in a structured scientific report, developing skills in realworld geophysical data analysis directly applicable to geoscience fieldwork and research support.
- **Relevant Project: Remote Sensing Elective; Lake Level & Algal Bloom Analysis (SWOT, Sentinel-1/2)**
  Processed SWOT and Sentinel-1 data to monitor lake-level and shoreline changes and used Sentinel-2 to calculate an algae index and track bloom intensity. Combined radar, altimetry and optical analysis using Python and GIS tools to assess drought impacts on water quality. This project strengthened my interest in radar missions such as BIOMASS and the use of P-band observations for environmental applications.
- **Relevant Project: Planetary Geomorphology Elective; Comparative Analysis of Titan vs Earth Hydrology.** Conducted a comparative geomorphological study of fluvial and lacustrine surface features on Titan and Earth, using Cassini-Huygens SAR imagery, near-infrared mapping, and altimetry data. Applied established Earth surface process principles, including dendritic drainage, meander geometry, delta formation, and lake-level dynamics, to interpret Titan's methane-cycle-driven landscapes. Strengthened understanding of radar remote sensing interpretation and geomorphological analysis in planetary contexts, directly relevant to TCD's research in terrestrial and non-terrestrial geomorphology.

BSc Physics with Astronomy, Dublin City University, Ireland **10/19 - 05/23**

- **Final GPA:** Upper Class Honours (2.1)
- **Key Modules:** Astrophysics & Cosmology, Astronomical Techniques, Applied Spectroscopy, Image Processing & Analysis, Computational Physics, Laboratory Reporting & Presentation, Advanced Programming, Physics Laboratory (I– III), Electromagnetism, Plasma Science and Technology.
- **Final Year Thesis**: Developed and validated a computational model of an Earth–Jupiter orbital transfer, using NASA JPL reference datasets to assess physical accuracy and efficiency. Focused on interpreting model behaviour and limitations rather than algorithmic optimisation.

VOLUNTARY EXPERIENCE

- Class Rep, MSc Space Science & Technology (09/25 - 09/26)
- Class Rep, BSc Physics with Astronomy (10/20 - 05/23)
- Treasurer, Amnesty International Society DCU. (09/19 - 05/21)
- Carer & Walker, DSPCA Animal Shelter. (06/17 - 09/18)

LANGUAGES

- English C2
- Irish C2
- French B1
- Dutch A1

SKILLS

TECHNICAL SKILLS

Remote Sensing (SAR & InSAR, SNAP, SWOT, QGIS & GIS & ArcGIS)
Python (NumPy, Pandas, Matplotlib, AstroPy, SciPy), MATLAB
Structured technical writing and reporting
CAD Design & 3D Printing
Electronics & Circuit Engineering
Spectroscopy & Optical Systems
Detector Characterisation, Calibration & Performance Analysis
Laboratory Experimentation & Measurement
Data Analysis, Modelling, Simulation & Visualisation (2 & 3D)
Basic Machine Learning, Data & Image Processing
Electromagnetism, Thermodynamics & Applied Physics

PROFESSIONAL

Team Leadership
Problem Solving
Content Structure & Organisation
Collaboration
Adaptability
Cross Disciplinary Communication
Time Management
Critical Thinking
Emotional Intelligence
Presentation & Networking
Personal Communication (Written & Oral)

ACHIEVEMENTS & CERTIFIFICATIONS

- Drone Flight License; Operator + A1/A3 certificate
- Full NDLS Driving License
- 'Space Mission Design Skills 2026, University of La Laguna Tenerife' Project Winner
- Only Irish university student to secure a European Space Agency internship for 2026
- 3rd Overall Class Rank (BSc Physics with Astronomy)
- Received recognition monetary scholarship award for achieving high Leaving Cert Marks
- Full Attendance Award for 6 years consecutive yearly attendance in Coláiste Íosagáin Secondary School.
- High Distinction on Undergraduate Thesis
- Team Sport Achievement; GAA All Ireland Football 2015 Winner

INTERESTS

- **Research Interests;** Geophysical research, Astronomical instrumentation, experimental observatory science.
- **Personal Interests;** 3D Printing, Drone Photography, Astrophotography, Scuba Diving, Sailing, Hiking.

References are available upon request.